



# XQ Secure Workspaces (iOS) —

## Executive Summary

### AI-Guided Governance

**Phased Scope:** File Vault + Secure Email + Secure Chat

**Core Addition:** AI-driven, on-device governance and policy automation (“at the edge”)

**Design Principle:** Security and compliance happen automatically, without user effort

**Platform:** iPad & iPhone

[XQ](#) Secure Workspaces is a privacy-first mobile security platform for iOS that combines secure file storage, protected email, private chat and AI-driven governance into a single, frictionless user experience. Built for both consumers and enterprises, the product automatically classifies, protects, and governs sensitive information directly on the device without requiring user training, complex setup, or constant IT oversight.

The platform extends XQ’s Zero Trust Data architecture to mobile environments, enabling organizations to secure documents, communications, and workflows while maintaining consumer-grade simplicity.

XQ Secure Workspaces is an iOS app that combines:

1. **Phase 1 : Secure File Vault**
2. **Phase 2 : Secure Email with protected attachments**
3. **Phase 3: Secure Chat**

### Core Value Proposition

XQ uniquely delivers a converged model of:

- **Immediate consumer adoption** with zero setup friction
- **Enterprise-grade governance** with policy-as-code and centralized control
- **Data-centric security enforcement**, where encryption, access control, and auditability are intrinsic properties of the data itself

This eliminates the traditional tradeoff between usability and security, enabling organizations to scale from individual usage to full enterprise deployment within a unified platform.





## Overview

The app functions as a secure container: files, emails and chats can be accessed within it, but cannot be exfiltrated from it. XQ holds no customer data at any point, and no content can be moved outside the app to third parties.

Enhanced with **AI-guided experience and governance at the edge**, the app automatically:

- Enable secure collaboration
- Prevents sensitive data leaving
- Enable secure collaboration
- Applies encryption, labeling, and access policies in real time

All enforcement remains aligned with XQ's **Zero Trust Data model**, ensuring compatibility across the XQ platform.

## Strategic Outcome + Market Opportunity

The XQ iOS Secure Workspace establishes a new category of mobile-first AI-driven, Zero Trust Data platforms by transforming the mobile device into a governed, encrypted data endpoint. The app enables secure data creation, sharing, and collaboration across consumer and enterprise environments while maintaining consistent enforcement of sovereignty, compliance, and access control policies.

The result is a unified system where data remains protected, portable, and policy-bound at all times, supporting both organic user adoption and mission-critical enterprise deployment at global scale.

### Market Opportunity

The convergence of AI adoption, mobile-first workflows, escalating cyber threats, and tightening global data sovereignty regulations has created a significant market gap between consumer productivity tools and enterprise-grade secure collaboration platforms. Organizations increasingly require solutions that can protect sensitive data beyond traditional network perimeters while enabling seamless user experiences across personal and corporate devices.

XQ is positioned to capture this emerging market by delivering a platform that combines Zero Trust security, AI-native workflows, encrypted collaboration, and policy-enforced data governance in a single mobile-centric architecture. The opportunity spans regulated industries, government and defense, critical infrastructure, global enterprises, and privacy-conscious consumers seeking secure digital workspaces that operate across distributed environments.

As enterprises transition toward AI-enabled operations and decentralized work models, the demand for persistent data protection and sovereign control at the endpoint level is expected to accelerate, creating a scalable platform opportunity for XQ to become foundational infrastructure for secure mobile collaboration and intelligent data exchange.





## CORE CAPABILITIES

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### **File & Email and Chat Connectivity**

Native connections to SharePoint and network drives. Files are streamed and accessed in-app — never stored on XQ infrastructure.

### **Secure Viewing & Containment**

In-app file viewing with active screenshot blocking and prevention of local download or export. Content cannot be moved outside the app or shared to third-party applications on the device.

### **AI enabled/supported Onboarding & Account Setup**

Streamlined first-run experience with automatic account creation. AI guides users through connecting to SharePoint or a network drive — detecting configuration, resolving common errors, and minimising IT involvement.

### **Offline Mode**

Users can designate files or folders for offline availability. Locally cached copies remain viewable and editable without a connection, with changes syncing back to source when connectivity is restored. Offline cache remains within the secure app container.

### **Sharing**

Sharing via SharePoint (leveraging its existing permissions and link model) and email. All sharing is controlled and auditable — files always resolve from the customer's own infrastructure.

### **Local File Import**

Users can bring in files from iOS local storage (Files app, camera roll, etc.). AI assists by suggesting destination folders, tagging content, and surfacing related files already in connected drives.

### **AI powered Document scanning**

Contextual analysis of file contents runs entirely on-device. No file data leaves the user's device — satisfying enterprise data residency requirements by design.

### **AI-Driven Policy Engine**

The platform applies configurable policies based on what the on-device AI identifies within files — restricting actions, triggering alerts, or enforcing access rules based on content classification.

### **Light Editing**

Scoped editing for text files, Word documents, and Excel spreadsheets. Changes save directly back to the source location — SharePoint or network drive.



# Product Definition



# Product Definition

XQ Secure Workspaces delivers effortless security through on-device intelligence.

Users work naturally with files and email while AI continuously:

- Detects sensitive information
- Applies protection policies
- Prevents data leakage
- Guides secure sharing
- Enforces governance automatically

Security becomes ambient, invisible, and continuous rather than manual and disruptive.

## Deployment Model Strategy

The architecture supports three distinct deployment modes:

- **Free Tier (Consumer / Local-First):** On-device encryption using Secure Enclave, no account required, zero infrastructure dependency
- **Enterprise SaaS:** XQ-managed platform with customer-controlled keys, policy engine enforcement, and full auditability

This structure enables bottom-up adoption while supporting top-down enterprise standardization without architectural change.





## Deployment Model Strategy

The platform supports three deployment models through a unified architecture.

### Free Tier — Consumer / Local-First

A privacy-first standalone deployment designed for individuals and small teams.

#### Characteristics

- No account required
  - Sharing / connected services
- Local-only operation
- On-device encryption using Apple Secure Enclave
- Zero infrastructure dependency
- AI governance fully local
- No cloud data exposure

#### Primary Value

Consumer-grade simplicity with automatic protection for sensitive files and communications.

### Enterprise SaaS

A managed enterprise deployment operated by XQ with customer-controlled governance.

#### Characteristics

- XQ-managed infrastructure
- Customer-controlled encryption keys
- Enterprise policy engine enforcement
- Full auditability and compliance visibility
- Passive enterprise identity integration
- Centralized administration

#### Primary Value

Enterprise-scale Zero Trust security with minimal operational friction.





## Core Product Components

### Secure File Vault

#### Phase 1

A protected workspace for storing, viewing, editing, and securely sharing files.

#### Core Capabilities

- Encrypted local file storage
- AI-powered content scanning and classification
- Automatic policy enforcement
- Secure sharing workflows
- Protected attachment handling
- Dynamic access control
- Policy-aware synchronization

#### Editing Experience

The product intentionally focuses on lightweight productivity rather than full office-suite replacement.

Users can:

- Open documents
- Make quick edits
- Save changes securely

After edits:

- AI automatically rescans content
- Policies update dynamically
- Classification remains current

Supported editing scenarios include:

- Microsoft Word documents
- Excel spreadsheets
- Common mobile productivity workflows





# Secure Email

## Phase 2

A secure email experience integrated directly into the protected workspace.

### Core Capabilities

- Email body scanning
- Attachment inspection
- Automatic encryption
- Secure attachment viewing
- AI-assisted risk detection
- Policy-driven message controls

### Intelligent Security Behavior

When sensitive information is detected, the system can:

- Recommend secure delivery methods
- Encrypt content automatically
- Restrict forwarding
- Prevent non-compliant sending
- Apply enterprise governance policies







# AI-Guided Governance Layer

## Phase 1

### Overview

AI-guided governance is the foundational intelligence layer of XQ Secure Workspaces.

The platform continuously analyzes content and context directly on-device to automate:

- Classification
- Encryption
- Access control
- Compliance enforcement
- Sharing governance

The user is never expected to manually manage security settings or compliance decisions.

### On-Device AI Architecture

#### Privacy-First Intelligence

AI models run locally by default.

No raw content leaves the device unless explicitly permitted by enterprise policy.

#### AI Analysis Capabilities

- Document content scanning
- Metadata inspection
- Email body analysis
- Attachment inspection
- Pattern detection
- Contextual classification
- Policy recommendation





## **Sensitive Data Detection**

The platform identifies:

- Personally Identifiable Information (PII)
- Protected Health Information (PHI)
- Controlled Unclassified Information (CUI)
- Financial records
- Compliance-sensitive content
- NIST

## **Dynamic Classification**

AI continuously classifies content using contextual risk analysis.

### **Example Classifications**

- Public
- Internal
- Confidential
- Restricted
- Custom

Classification automatically drives downstream policy enforcement.





# Policy Automation

## Consumer Policy Automation

For consumer deployments, governance emphasizes simplicity and passive protection.

### Automated Behaviors

- Encrypt all files and emails automatically
- Warn before risky sharing actions
- Recommend secure sharing methods
- Apply local protection silently
- Support secure editing workflows
- Enable file and email revocation

The experience remains lightweight and minimally intrusive.

## Enterprise Policy Automation

Enterprise deployments introduce centralized governance and dynamic enforcement.

### Automated Behaviors

- Restrict external sharing with individuals and groups
- Apply geofencing controls
- Enforce view-only access
- Trigger audit logging
- Apply group-based permissions
- Enforce attribute-based access control (ABAC)
- Prevent policy violations before transmission
- Dynamic policy generation sent to backend

Policies adapt dynamically based on:

- User identity
- Device posture
- Location
- Content sensitivity
- Organizational policy





## Real-Time Security Decisioning

| Event           | Automated Action                                                   |
|-----------------|--------------------------------------------------------------------|
| File imported   | AI scans → classifies → encrypts                                   |
| Email composed  | AI scans → evaluates risk → applies policy->conditionally encrypts |
| File shared     | AI validates recipient, context, and sensitivity, group            |
| Document edited | AI rescans and updates protections                                 |
| Policy changes  | Access rules update dynamically                                    |

## Security Model

### Secure Container Architecture

XQ Secure Workspaces is fundamentally a secure container platform.

Content enters only through controlled pathways:

- Microsoft SharePoint
- Local device storage
- Enterprise repositories
- Approved integrations

Data cannot leave the secure workspace except through explicit, auditable sharing workflows.

## Data Leakage Prevention

Security protections are foundational rather than optional.





## Built-In Controls

- Screenshot blocking where iOS permits
- Sensitive-content watermarking
- Copy/export prevention
- Share-sheet restrictions
- Background blur protection
- Secure attachment handling

## AI-Enhanced Protection

High-risk content can trigger stricter runtime protections automatically.

# Sharing Experience

## Secure Sharing Workflow

The sharing experience is designed to remain simple while enforcing governance automatically.

### User Flow

1. User selects “Share Securely”
2. Chooses recipient
3. AI evaluates content sensitivity
4. Appropriate protections apply automatically

## AI-Assisted Decisions

The system can:

- **Recommend links vs direct attachments**
- Warn on external recipients
- Detect risky destinations
- Suggest stronger controls
- Restrict sharing to only authorized parties





## Enterprise Enforcement

Organizations can automatically apply:

- Expiration policies
- View-only access
- Geofencing
- Access revocation
- Server-side offloading
- Audit tracking
- Labels and classification
- Policy generation

# Onboarding Experience

## Consumer Onboarding

Designed for immediate use with minimal friction.

### Characteristics

- Instant access
- AI guided onboarding and customization
- No mandatory login
  - Login only required to connect services
  - Optionally User can login to create account
- No required training
- Silent AI governance activation
- Automatic protection on first use

The first imported file automatically triggers:

- Classification
- Encryption
- Governance enforcement





## Enterprise Upgrade Path

Consumer deployments can transition seamlessly into enterprise governance.

### Upgrade Behaviors

- Existing files are reclassified automatically
- Enterprise policies apply retroactively
- No workflow disruption
- No user retraining required

## Interoperability

### Native Integrations

Supported ecosystem integrations include:

- Microsoft SharePoint
- Apple Files (Optional)
- Outlook
- Gmail
- External secure file links

Potential future integrations:

- Google Workspace
- Expanded Microsoft suite
- Additional enterprise repositories





## XQ Ecosystem Integration

AI-generated classifications become reusable policy signals across the broader XQ platform.

This enables:

- Consistent policy enforcement
- Shared governance intelligence
- Unified Zero Trust workflows
- Cross-platform interoperability
  - Works with desktop app

## User Experience Philosophy

### Consumer-Grade Simplicity

The interface is intentionally minimal.

#### UX Principles

- No policy dashboards for standard users
- No manual classification requirements
- Minimal prompts
- Risk-based intervention only
- Native iOS interaction patterns

#### User Signals

Security visibility remains lightweight through:

- Small sensitivity labels
- Contextual warnings
- Subtle policy indicators







# Enterprise Administration

## Administrative Controls

Enterprise administrators can:

- Define classification schemas
- Configure AI sensitivity thresholds
- Override AI decisions
- Monitor policy violations
- Audit data movement
- Configure governance models
- User access escalation resolution

## Risk Management

| Risk                      | Mitigation                                |
|---------------------------|-------------------------------------------|
| AI false positives        | User overrides + model refinement         |
| Device performance impact | Lightweight on-device models              |
| Privacy concerns          | Local processing by default               |
| Workflow disruption       | Progressive enforcement and soft warnings |

## Success Metrics

The platform measures effectiveness through:

- Auto-classification accuracy
- Risky-share prevention rates
- Reduction in data leakage incidents
- User effort per workflow
- Enterprise compliance adherence
- Policy enforcement effectiveness





# Final Positioning

## XQ Secure Workspaces

**Effortless security powered by AI-driven governance at the edge.**

The platform combines:

- Secure file storage
- Protected communications
- Real-time AI governance
- Zero Trust enforcement
- Consumer-grade usability

Users work naturally while AI continuously protects data, enforces policy, and prevents risk — without introducing friction or operational complexity.

Should understand if recipient has XQ account for encrypted to encrypted email and file sharing?



# Feature Specification



# XQ Secure Workspaces AI

## Feature Specification

### 1. Product Definition

#### Overview

XQ Secure Workspaces is an AI-native secure workspace platform providing lightweight productivity, and autonomous policy enforcement for iOS and iPadOS that combines:

- Secure file management
- Secure communications
- AI-driven governance
- Persistent Zero Trust data protection
- Enterprise policy enforcement
- Local-first encrypted storage
- Ambient security automation

The application is designed to make:

Enterprise-grade security feel invisible and consumer-simple.





## 2. Core Product Principles

| Principle              | Description                                        |
|------------------------|----------------------------------------------------|
| Ambient Security       | Security operates silently in background           |
| AI-First Interaction   | Natural language becomes primary interaction model |
| Local-First Privacy    | Sensitive operations remain on-device              |
| Calm UX                | Reduce cognitive overload                          |
| Explainable Governance | AI decisions are human-readable                    |
| Adaptive Enforcement   | Runtime security changes dynamically               |
| Unified Workspace      | Files, messaging, AI, governance converge          |
| Minimal Friction       | Security should not interrupt workflows            |

## 3. Platform Scope

### Supported Platforms

- iPhone
- iPad

### Initial Operating System Support

- iOS (latest two major releases minimum)
  - iOS/iPadOS 17

### Future Expansion

- macOS
- Android
- Desktop secure workspace clients





## 4. User Types

### 4.1 Personal / Consumer User

#### Goals

- Protect personal files
- Share securely
- Operate offline
- Avoid cloud dependency

#### Product Behaviors

- No account required
- Local encrypted vault
- On-device AI only
- Optional iCloud integration

### 4.2 Enterprise User

#### Goals

- Access governed SharePoint and other connected services content securely
- Maintain compliance automatically
- Collaborate securely
- Minimize security complexity

#### Product Behaviors

- SSO authentication
- Automatic repository mounting
- AI policy enforcement
- Continuous governance





## 4.3 Enterprise Administrator

### Goals

- Define policies
- Control sharing
- Audit activity
- Tune AI governance

### Product Behaviors

- Embedded admin controls
- Dynamic policy publishing
- Runtime enforcement tuning
- Tenant governance visibility

## 5. Navigation Architecture

### 5.1 Primary Navigation

The product now uses a simplified:

#### 3-tab navigation model

| Tab      | Screen                       | Purpose                            |
|----------|------------------------------|------------------------------------|
| Files    | <code>s-file-browser</code>  | Protected repositories and vault   |
| Messages | <code>s-email-inbox</code>   | Secure communications and sharing  |
| Now      | <code>s-notifications</code> | AI dashboard, alerts, intelligence |





## 5.2 Profile Architecture

The old Settings tab has been replaced with:

### Persistent Profile Command Center

Accessible via:

- Top-right avatar (BW)
- Available globally across top-level screens

The Profile screen centralizes:

- Identity
- Security
- Notifications
- Integrations
- Workspace management
- Device management
- Billing
- Enterprise administration

## 6. “Now” Dashboard Architecture (s-notifications)

### Overview

Now replaces the traditional Home screen.

Purpose:

- Operational intelligence surface
- AI interaction hub
- Security awareness dashboard
- Action prioritization engine







## 6.1 Now Sub-Tabs

| Sub-Tab  | Purpose                       |
|----------|-------------------------------|
| Overview | Workspace operational summary |
| Ask      | AI conversational assistant   |
| Activity | File/share/access events      |
| Security | Threats and policy violations |

## 6.2 Overview Sub-Tab

### Functional Components

#### Trust Status Pill

Displays:

- Workspace health
- Scan recency
- Protected file counts

Example:

None

Workspace protected · scanned just now

### Today Feed





Chronological event timeline:

- Secure file imports
- Share events
- Vault scans
- Policy updates
- AI actions

## Needs Your Attention

Single-row actionable cards:

- Expiring shares
- External access
- Received secure files
- Risk escalations

## Suggested Actions

AI-generated operational suggestions:

- Apply labels
- Cleanup stale shares
- Restrict permissions
- Reclassify files

## 6.3 Ask Sub-Tab (AI Assistant)





## Purpose

Primary AI interaction surface.

This replaces:

- Dedicated AI tab
- Traditional search workflows
- Semantic search screens

## AI Capabilities

- Enable workflow
- Enforce policies
- Make policy recommendations

## File Intelligence

- “Which files contain PHI?”
- “Find contracts shared externally”

## Security Analysis

- “Summarize my security posture”
- “Show high-risk shares”

## Governance Explanations

- “Why is this file restricted?”
- “Who can access this document?”





## Runtime Characteristics

| Capability | Behavior                                                            |
|------------|---------------------------------------------------------------------|
| Inference  | On-device                                                           |
| Models     | CoreML / ONNX →                                                     |
| Input      | Natural language                                                    |
| Search     | Semantic embeddings                                                 |
| Security   | Local-only by default. Keys held with XQ. Offline access supported. |

## 7. Files Architecture

### 7.1 File Browser & Editor (s-file-browser)

#### Simplified Browsing Model

##### Sub-tabs

| Section | Purpose                 |
|---------|-------------------------|
| Folders | Structured repositories |
| Recent  | Chronological access    |
| Shared  | Shared content          |
| Vault   | Local encrypted storage |

- Open question: should AI reorganize how your files are stored to organize them for you?





## 4.9 Light Editing

### Description

The platform supports lightweight productivity workflows without replacing full desktop office suites.

### Supported Editing

- Text files
- Microsoft Word documents
- Excel spreadsheets

### Functional Requirements

Users can:

- Open documents
- Make lightweight edits
- Save directly to source repository

### Post-Edit Processing

After save:

- AI rescans content
- Classification updates dynamically
- Policies re-evaluate automatically

### DOCUMENT EDITING CAPABILITIES

Add support for:

- - Native document editing





- - Real-time collaborative editing
- - Offline editing with secure sync
- - Version history
- - Restore previous versions
- - Secure autosave
- - Inline commenting
- - Suggested edits
- - Track changes
- - Approval workflows
- - AI-assisted editing
- - Rich text formatting
- - Embedded metadata editing
- - Secure annotation support

Supported file types:

- - Text documents
- - Markdown
- - Office documents
- - PDFs with annotations
- - Structured enterprise forms
- - AI-generated reports

## EDITABILITY STATES





Every document and shared file must visually indicate its access/edit state.

Required document states:

- - Editable
- - View Only
- - Comment Only
- - Pending Approval
- - Restricted
- - Expired Access
- - External Shared
- - Local Only
- - AI Restricted

## 7.2 AI Search Routing

Search bars no longer perform direct filtering.

Instead:

```
None
Search Input
  ↓
Now → Ask
  ↓
AI Semantic Query
  ↓
Result Actions
```

## 7.3 AI Organize (**s-ai-organize**)





## AI Features

- Similarity clustering
- Suggested foldering
- Sensitive content grouping
- Duplicate reduction
- Workspace cleanup recommendations

## 7.4 Secure File Viewer & Editor / (s-file-viewer)

### Runtime Protection Environment

#### Security Controls

- Screenshot blocking
- Screen recording detection
- Background blur
- Secure rendering container
- Clipboard restrictions
- Dynamic watermarking

Light editing for allowed files and file types:

- Docx
- Txt
- Excel
- (Future) PPTX

## 7.5 Security Intelligence Panel (SIP)







## Displays

- Classification
- Threat scan results
- Risk indicators
- Policy status
- Data lineage

Minimized by default in:

None

Display Mode = Minimized

## 7.6 Data Lineage (**s-data-lineage**)

Tracks:

- Origin
- Repository history
- Sharing history
- Access history
- Geographic access
- AI classification evolution

## 7.7 AI Risk Dashboard (**s-file-risk-dashboard**)





## AI Detection Categories

| Risk                    | Example             |
|-------------------------|---------------------|
| Credential exposure     | Secrets/API keys    |
| PHI exposure            | Medical records     |
| Permission drift        | Oversharing         |
| Classification mismatch | Incorrect labeling  |
| Sensitive duplication   | Uncontrolled copies |

## 8. Secure Messaging Architecture

### 8.1 Messages Tab (**s-email-inbox**)

Messages now merges:

- Secure email
- Sharing awareness
- AI compliance scanning
- Ai composition

### 8.2 Inbox Features





## Features

- Threaded view
- Classification badges
- Risk indicators
- Secure attachment rendering

Search routes to:

None

Now → Ask

## 8.3 AI Threat Analysis (s-phishing-alert)

AI detects:

- Phishing
- Credential harvesting
- Suspicious domains
- Tone anomalies
- Impersonation attempts

## 8.4 Secure Compose (s-email-compose)

### AI Pre-Send Review





Evaluates:

- Recipient trust
- PHI/PII exposure
- Sensitive commitments
- Export-controlled content
- Tone/sentiment risk

## Automated Enforcement

| Condition            | Enforcement             |
|----------------------|-------------------------|
| External PHI         | Force encryption        |
| High-risk recipient  | Require confirmation    |
| Policy violation     | Block send              |
| Sensitive attachment | Secure share conversion |

## 9. Sharing Architecture

### 9.1 Sharing Model





Sharing is now:

## AI-governed by default

Users typically do not manually configure security.

## AI Determines

| Control          | Dynamic Behavior |
|------------------|------------------|
| View-only        | Risk adaptive    |
| Expiration       | AI recommended   |
| Download rights  | Contextual       |
| Watermarking     | Dynamic          |
| Geofencing       | Policy-aware     |
| MFA requirements | Risk-based       |

## 9.2 Received Shares (**s-received-share**)

Requirements:





- XQ app required
- Policy-aware decryption
- Runtime validation
- Continuous authorization

## 9.3 Workgroup Invites (**s-group-invite**)

(These map back to workspaces and Groups in XQ admin)

Invitations include:

- Policy previews
- Workspace classification
- Governance inheritance
- Risk indicators

# 10. Workgroup Architecture

## 10.1 Workspace Model

A user may belong to:

- Enterprise workspaces
- Departmental workspaces
- Personal vaults
- Shared collaboration groups

## 10.2 Workspace Management

Now consolidated under:





None

[Profile](#) → [Workspace](#)

## Workspace Metadata

| Metadata    | Example         |
|-------------|-----------------|
| Tenant ID   | Enterprise GUID |
| Policy tier | HIPAA / SOC2    |
| Residency   | US-West         |
| Group count | Dynamic         |
| Memberships | Role aware      |

## 11. Profile Command Center (**s-profile**)

### Overview

Profile replaces:

- Settings
- Account management
- Admin console
- Security center

Single vertically-scrolling command surface.

### 11.1 Identity Card

Displays:

- Avatar





- Name
- Organization
- Role
- Verified status
- Admin status
- Last login metadata

## 11.2 Security Health Card

Mini-tiles:

- MFA enabled
- Active sessions
- Policy compliant
- Encryption status

## 11.3 Sticky Navigation Chips

Persistent subsection navigation:

- Smooth scroll
- State persistence
- Last-section restoration

Stored via:

None

`xq.profileSection`

## 11.4 Profile Sections

| Section | Purpose              |
|---------|----------------------|
| General | Identity/preferences |







|               |                                                                                    |
|---------------|------------------------------------------------------------------------------------|
| Security      | MFA/device/runtime                                                                 |
| Notifications | Alert channels                                                                     |
| Integrations  | Connected services                                                                 |
| Workspace     | Workgroups/tenant                                                                  |
| Devices       | Session management                                                                 |
| Billing       | Subscription/license (NOT sure what features the premium Consumer model will have) |
| Admin         | Enterprise governance                                                              |

## 12. Integrations Architecture

### 12.1 Supported Integrations

| Integration  | Status                   |
|--------------|--------------------------|
| SharePoint   | Native (XQ supports)     |
| OneDrive     | Native (XQ supports)     |
| Entra ID     | Native (XQ supports)     |
| Outlook 365  | Native (XQ supports)     |
| Slack        | Connected (Low priority) |
| Okta         | Native (XQ supports)     |
| Gmail        | Planned (XQ supports)    |
| Google Drive | Planned                  |





## 12.2 Integration Interface Model

No direct API calls permitted.

All integrations use:

Swift

```
protocol ExternalIntegrationProvider {  
    func authenticate()  
    func sync()  
    func query()  
    func revoke()  
}
```

## 13. Enterprise Admin Architecture

### 13.1 Admin Section (**Profile** → **Admin**)





Embedded governance experience.

# Governance Controls

## Classification Rules

- PHI auto-detection
- PII auto-detection
- Financial data scanning
- INcluded frameworks selected by administrator

## Share Enforcement

- Follow Policies
- Maximum share expiry
- Require XQ app
- Screenshot enforcement

## AI Governance Gates

- Cloud AI enable/disable
- Local-only enforcement
- Sensitive content routing
- AI confidence thresholds

## Dynamic Classification System

## Classification Labels

Examples include:

- Public
- Internal
- Confidential





- Restricted
- \*custom

## Requirements

- Continuous reclassification
- Context-aware sensitivity evaluation
- Multi-label support

## Real-Time Security Decisioning

| Event           | Automated Action                      |
|-----------------|---------------------------------------|
| File imported   | Scan → classify → encrypt             |
| Email composed  | Scan → risk evaluate → enforce policy |
| File shared     | Validate recipient/context            |
| Document edited | Rescan + update protections           |
| Policy changed  | Recalculate access rules              |

## 13.2 Policy Publishing

### States

- Draft
- Signed
- Published





- Revoked

Admin actions:

- Publish
- Discard
- Export audit logs

## 14. Security Architecture

### 14.1 Core Security Model

| Capability        | Implementation      |
|-------------------|---------------------|
| Encryption        | AES-256-GCM         |
| Key custody       | Secure Enclave      |
| Runtime integrity | Jailbreak detection |
| Share security    | Zero-knowledge keys |
| Offline support   | Secure replay queue |

### 14.2 Runtime Protection

#### Dynamic Controls

- Screenshot blocking (Very important )
- Background obfuscation
- Clipboard restriction
- Runtime watermarking





- Session timeout escalation

## 14.3 Revocation Architecture

Revocation propagates to:

- Offline cache
- Shared workspace access
- Device sessions
- Temporary keys

## 15. AI Architecture

### 15.1 AI Runtime Model

| Capability      | Runtime |
|-----------------|---------|
| Classification  | Local   |
| Semantic search | Local   |





|                      |       |
|----------------------|-------|
| Risk analysis        | Local |
| OCR                  | Local |
| Governance reasoning | Local |

## 15.2 AI Model Architecture

Supports interchangeable:

- CoreML models
- ONNX models
- Enterprise models
- Future cloud orchestration

All AI accessed through:

Swift  
`protocol AIProvider`

## 16. Persistence & Routing Architecture

### 16.1 Route Consolidation

Legacy Route      New Route





s-home

s-notificatio  
ns

s-settings

s-profile

s-ai-assistant

Now → Ask

s-semantic-sear  
ch

Now → Ask

## 16.2 Local Persistence Keys

| Key               | Purpose                         |
|-------------------|---------------------------------|
| xq.notifMode      | Expanded/minimized display mode |
| xq.profileSection | Last profile subsection         |
| xq.onbPath        | Onboarding mode                 |

## 17. Animation & Motion System

Every UX element must animate.







# Motion Categories

| Interaction    | Animation             |
|----------------|-----------------------|
| Entry          | Fade + motion         |
| Exit           | Scale + fade          |
| Tap            | Spring response       |
| Risk alert     | Contextual emphasis   |
| Secure actions | Reinforced transition |

## UX Goals

| Metric            | Target |
|-------------------|--------|
| FPS               | 60     |
| Cold boot         | <3s    |
| Local AI response | <300ms |
| Cached file open  | <1s    |

## 18. Phase 1 Constraints

Phase 1 intentionally excludes:

- Real-time collaborative editing
- macOS/Windows/Android clients
- Separate admin web console





- XQ will have to do this part I think
- Non-XQ vault backups
- Generative document drafting

## 19. Strategic Product Positioning

XQ Secure Workspaces is evolving into:

An AI-native sovereign workspace where files, communications, governance, and security become ambient services rather than separate tools.

The platform differentiates through:

- On-device AI governance
- Zero Trust Data protection
- AI-driven policy orchestration
- Local-first secure intelligence
- Consumer-grade enterprise UX
- Dynamic runtime protection
- Explainable security automation

## APPENDIX

## XQ Secure Workspace





# AI Composition Assistance — Feature Specification

## 1. Overview

AI Composition Assistance introduces embedded, policy-aware AI capabilities across Secure Email and Files within the XQ Secure Workspace platform.

The feature enables users to draft, summarize, rewrite, analyze, classify, and organize communications and documents while maintaining Zero Trust Data governance, enterprise compliance, sovereignty enforcement, and full auditability.

The AI system must operate as an ambient productivity layer integrated directly into workflows rather than as a standalone chatbot interface.

## 2. Goals

### Primary Goals

- Improve user productivity
- Reduce friction in secure communications
- Accelerate document workflows
- Enhance secure collaboration
- Preserve Zero Trust governance
- Support tactical edge operations
- Enable sovereign and air-gapped deployments

### Non-Goals

- Consumer-style open AI chat experience
- Unrestricted internet-connected AI
- Uncontrolled model access
- Direct client-to-model communication





- Non-auditable AI actions

## 3. Product Principles

| Principle              | Description                              |
|------------------------|------------------------------------------|
| Embedded AI            | AI appears contextually inside workflows |
| Zero Trust First       | Every AI action is policy evaluated      |
| Human-in-Control       | Users remain decision makers             |
| Transparent Governance | AI actions explainable and auditable     |
| Minimal Friction       | Lightweight interactions preferred       |
| Sovereign Ready        | Supports local, private, and edge AI     |
| Secure by Default      | AI inherits existing data protections    |

## 4. Core Functional Areas

4.1 Email AI Assistance

4.2 Files AI Assistance

4.3 AI Governance

4.4 AI Orchestration Services

4.5 Secure Retrieval + Context Engine

4.6 Offline + Edge AI Support





## 5. Email AI Assistance

### 5.1 Compose Assistance

#### Features

- Inline predictive drafting
- Ghost-text suggestions
- Context-aware sentence completion
- AI-generated subject lines
- Smart greeting/signature suggestions
- Suggested recipients
- Attachment-aware drafting
- Workspace-aware drafting

#### User Actions

| Action        | Description                           |
|---------------|---------------------------------------|
| Help Me Write | Opens AI drafting assistance          |
| Rewrite       | Rewrites selected content             |
| Expand        | Expands brief text into full response |
| Summarize     | Summarizes selected thread            |
| Extract Tasks | Generates actionable tasks            |
| Translate     | Converts content language             |
| Adjust Tone   | Modifies communication style          |

### 5.2 Rewrite Modes

| Mode         | Purpose                |
|--------------|------------------------|
| Professional | Formal enterprise tone |





|                    |                            |
|--------------------|----------------------------|
| Concise            | Shortened response         |
| Executive          | Leadership-focused summary |
| Friendly           | Softer conversational tone |
| Technical          | Technical precision        |
| Compliance<br>Safe | Removes risky language     |

## 5.3 Email Summarization

### Capabilities

- Long-thread summarization
- Meeting recap generation
- Action item extraction
- Risk identification
- Timeline extraction
- Decision summaries

### Output Types

- Bullet summaries
- Executive summaries
- Action/task lists
- Follow-up drafts

## 5.4 Secure Send Intelligence

### AI Security Checks





Before sending:

- Detect sensitive data
- Detect regulated content
- Detect external recipients
- Validate attachment policies
- Check label inheritance
- Check sovereignty restrictions
- Evaluate DLP policy

## AI Recommendations

| Recommendation     | Example                        |
|--------------------|--------------------------------|
| Add Classification | "Recommend Restricted label"   |
| Restrict Sharing   | "External sharing blocked"     |
| Encrypt Attachment | "Sensitive file detected"      |
| Remove Recipient   | "Unauthorized external domain" |

## 6. Files AI Assistance

### 6.1 AI Document Creation





## Features

- Generate from prompt
- Smart templates
- Secure report generation
- Structured document creation
- Meeting note generation
- Policy-driven templates

## 6.2 Document Intelligence

### Features

- Executive summaries
- Auto-tagging
- Metadata generation
- Risk extraction
- Task extraction
- Compliance review
- Redaction recommendations
- Duplicate detection
- Knowledge extraction

## 6.3 Rewrite + Editing

### Features

- Rewrite content
- Simplify language







- Technical enhancement
- Formatting improvements
- Policy-safe rewrite
- Generate executive version
- Generate public-safe version

## 6.4 AI Knowledge Interaction

### Capabilities

- Contextual Q&A
- Multi-document synthesis
- Workspace-aware retrieval
- Authorized-only semantic search
- Enterprise knowledge summarization

### Restrictions

- No unauthorized retrieval
- No cross-tenant leakage
- No hidden context exposure
- No bypassing labels or RBAC

## 7. AI Governance Framework

### 7.1 Governance Principles

Principle

Enforcement





|                   |                               |
|-------------------|-------------------------------|
| Zero Trust        | Policy check before inference |
| Least Privilege   | Authorized context only       |
| Data Sovereignty  | Region-aware routing          |
| Auditability      | Full AI logging               |
| Explainability    | Explain recommendations       |
| Label Persistence | Outputs inherit labels        |

## 7.2 Audit Logging

### Logged Events

- User prompts
- Retrieved context
- Model selected
- Policy decisions
- Generated outputs
- Rewrite operations
- User approvals
- Blocked actions
- Label assignments

## 7.3 Explainability

Users must see:

- Why AI made recommendation
- Why action blocked
- Which policies applied





- Classification inheritance
- Confidence scoring

## 8. AI Classification Integration

### AI Label Awareness

| Label        | AI Behavior                    |
|--------------|--------------------------------|
| Restricted   | External inference blocked     |
| Confidential | Policy-controlled inference    |
| Internal     | Standard enterprise inference  |
| Public       | Broad AI operations allowed    |
| Local Only   | Device-only inference required |

## 9. AI Orchestration Architecture

### 9.1 Architecture Overview

All AI operations route through the XQ AI Orchestration Layer.

No direct model access from clients.





## Responsibilities

- Policy enforcement
- Model routing
- Prompt enrichment
- Retrieval orchestration
- Audit logging
- Streaming management
- Token governance
- Sovereignty routing

## 9.2 Supported Model Providers

| Provider Type       | Supported |
|---------------------|-----------|
| OpenAI              | Yes       |
| Azure OpenAI        | Yes       |
| Anthropic           | Yes       |
| Local LLMs          | Yes       |
| Air-Gapped Models   | Yes       |
| Sovereign Providers | Yes       |

# 10. Retrieval-Augmented Generation (RAG)

## Secure Retrieval Pipeline

### Workflow

1. User submits request





2. Identity validated
3. Policies evaluated
4. Authorized content retrieved
5. Labels filtered
6. Context sanitized
7. Prompt enriched
8. Model inference executed
9. Output policy validated
10. Response streamed

## 11. Offline + Tactical Edge AI

### Requirements

- Local inference support
- Sync queue architecture
- Delayed telemetry synchronization
- Edge-optimized models
- Intermittent connectivity support
- Air-gapped operation

## 12. User Experience Requirements

### 12.1 Email UX

#### Components

- AI toolbar above keyboard
- Inline suggestion chips





- Ghost-text drafting
- Streaming text insertion
- Secure warning banners
- Expandable AI panel

## 12.2 Files UX

### Components

- AI floating action button
- Inline insight cards
- Metadata chips
- Split-view AI assistant
- Embedded summaries
- Suggested actions panel

## 13. iPad Experience

### Requirements

- Multi-pane layouts
- Split-screen AI assistant
- Drag/drop AI actions
- Apple Pencil annotation support
- Keyboard shortcut integration

## 14. Backend Services

**Service**

**Responsibility**





|                   |                              |
|-------------------|------------------------------|
| AI Gateway        | Central orchestration        |
| Policy Engine     | Governance enforcement       |
| Retrieval Engine  | Authorized context retrieval |
| Label Service     | Classification management    |
| Audit Service     | AI logging                   |
| Model Router      | Model selection              |
| Prompt Service    | Prompt enrichment            |
| Streaming Service | Real-time output delivery    |

## 15. APIs

### Core APIs

| API                   | Purpose          |
|-----------------------|------------------|
| /ai/compose           | Draft generation |
| /ai/rewrite           | Rewrite content  |
| /ai/summarize         | Summarization    |
| /ai/classify          | Classification   |
| /ai/retrieve          | Secure retrieval |
| /ai/policies/evaluate | Policy checks    |
| /ai/audit/log         | Logging          |

## 16. Security Requirements





## Mandatory Controls

- TLS everywhere
- Tenant isolation
- Encrypted inference payloads
- Hardware-backed key protection
- Secure enclave support
- Policy enforcement pre/post inference
- Output sanitization
- Data minimization

## 17. Performance Targets

| Metric                  | Target         |
|-------------------------|----------------|
| Initial AI Response     | < 2<br>seconds |
| Streaming Latency       | < 300ms        |
| Summarization Time      | < 5<br>seconds |
| Rewrite Action          | < 2<br>seconds |
| Local Inference Startup | < 1 second     |

## 18. Administrative Controls

### Admin Features

- Allowed model selection
- AI capability enablement
- Label-specific restrictions
- Prompt retention policies
- Audit export







- Token limits
- Department-level controls
- Sovereignty routing rules

## 19. Compliance Support

### Supported Compliance Targets

- HIPAA
- GDPR
- CJIS
- ITAR
- FedRAMP
- CMMC
- NIST 800-207
- DoD Zero Trust

## 20. Success Criteria

| Success Metric              | Goal                 |
|-----------------------------|----------------------|
| Reduction in Draft Time     | >40%                 |
| Policy Violations Prevented | Measurable reduction |
| Unauthorized Data Exposure  | Zero                 |
| AI Response Satisfaction    | >90%                 |
| Governance Audit Coverage   | 100%                 |



# Application Specification



# XQ Secure Workspaces — Revised Application Specification

## Unified Cross-Platform Secure AI Workspace Platform

### 1. Product Vision

XQ Secure Workspaces is a unified Zero Trust secure productivity environment that combines:

- Secure files
- Secure email
- AI-assisted governance
- Enterprise repository access
- Offline-first secure workspaces
- Ambient compliance enforcement
- AI-driven classification and risk analysis

The platform operates as a secure runtime layer across:

- iPhone
- iPad
- Desktop/Web
- Future secure edge devices

Core design principles:

- Minimal cognitive load
- Local-first security
- Invisible governance
- Adaptive workspace layouts
- Touch-first interaction
- Enterprise-grade Zero Trust protections
- AI-assisted productivity without overwhelming the user





## 2. Platform Architecture

### Supported Modes

The application dynamically supports:

| Mode             | Purpose                               |
|------------------|---------------------------------------|
| Mobile Mode      | Minimal focused workflows             |
| iPad Mode        | Multi-pane productivity workspace     |
| Desktop/Web Mode | Full enterprise operational workspace |

### Dynamic Mode Switching

Users and developers can instantly switch between:

- Mobile
- iPad
- Desktop

Requirements:

- Preserve navigation state
- Preserve open files and drafts
- No application reload
- Responsive component re-rendering
- Persist mode preference across sessions



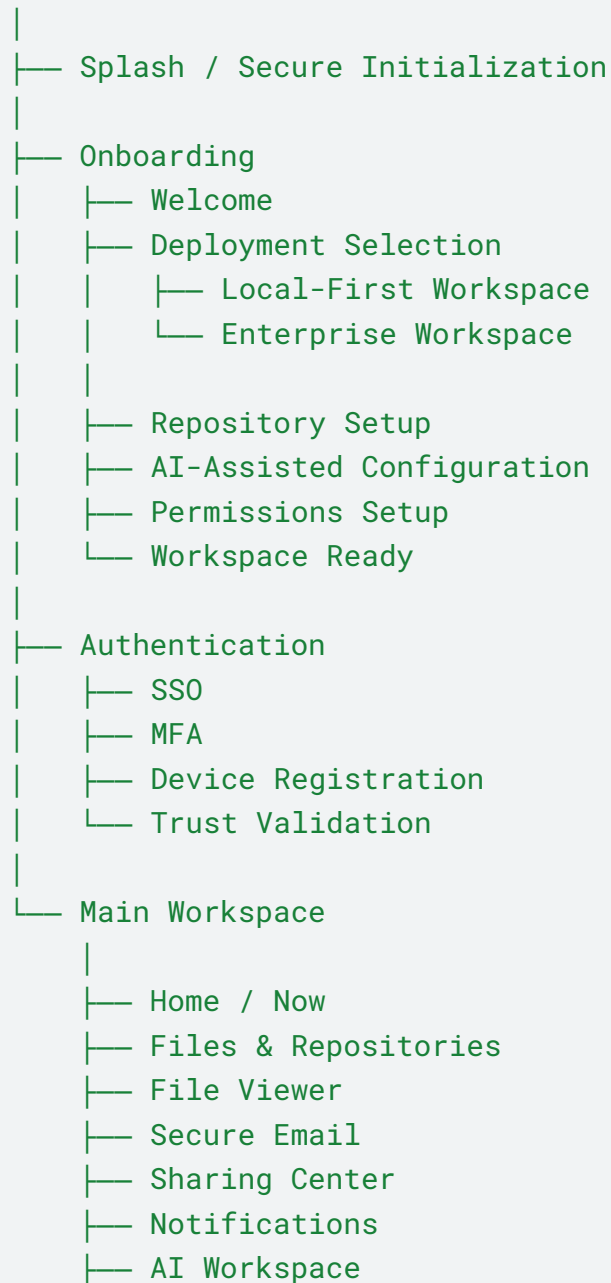


## 3. Global Application Structure

### Launch Experience

None

#### Launch





Enterprise Governance  
Settings

## 4. Primary Navigation Model

### Mobile Navigation

Bottom Tab Bar:

- Now
- Files
- Email
- Profile
  - Settings

### iPad Navigation

Adaptive Sidebar + Multi-Column Layout:

- Persistent navigation sidebar
- Split-pane content
- Optional detail inspector panel
- Slide-over contextual panels

### Desktop Navigation

Enterprise Workspace Layout:

- Left global navigation
- Center workspace
- Right AI/governance/context panel
- Multi-window support





## 5. Home / “Now” Workspace

### Purpose

The “Now” workspace provides:

- Immediate awareness
- Critical actions
- AI-prioritized productivity
- Lightweight governance visibility

### Layout Behavior

#### Mobile

- Vertical card feed

#### iPad

- Adaptive modular dashboard
- Multi-column cards
- Expandable sections

#### Desktop

- Configurable operational workspace

### Sections

#### Recent Files

- Continue working
- Recent edits
- Recent shares

#### Suggested Files

AI-generated based on:

- Context





- Projects
- Collaboration
- Time relevance

## Offline Workspace

- Available offline files
- Sync status
- Pending sync conflicts

## Risk Notifications

Examples:

- Sensitive file externally shared
- Unclassified document
- High-risk email detected

## Quick Actions

- Import file
- Secure share
- Compose secure email
- Scan document
- Launch AI workspace







## 6. Files & Repository Workspace

### Purpose

Unified secure access layer across:

- SharePoint
- SMB/Network drives
- Cloud repositories
- Local Secure Vault
- Future enterprise connectors

## 7. Repository Browser Specification

### Core Features

#### Repository Switcher

- Multi-repository support
- AI-suggested repositories
- Connection health indicators

#### File Navigation

- Folder hierarchy
- Grid/list views
- Smart search
- Classification filters

#### File Indicators

- Sensitivity labels
- Sync status
- Share status
- Offline availability
- Governance enforcement indicators





The colored stripe on the left edge of each folder card (.fc-edge) is the folder's **classification level**. There are five colors, each mapping to a sensitivity tier defined in the design tokens:

| Color                                                                                      | Class          | Meaning                                                                                                                                                                     | Where it appears                                   |
|--------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
|  Red      | fc-edg<br>-res | Restricted — highest sensitivity. PHI, PII, regulated data. Sharing is blocked or heavily gated.                                                                            | Legal, Human Resources, Patient Records            |
|  Orange   | fc-edg<br>-con | Confidential — sensitive business data. Internal default, external share requires policy approval.                                                                          | Finance, Finance & Legal                           |
|  Blue     | fc-edg<br>-int | Internal — workforce-only. Routine internal docs.                                                                                                                           | Engineering, Shared                                |
|  Green    | fc-edg<br>-pub | Public — no restriction. Marketing collateral, public docs.                                                                                                                 | Rarely used on folders; common on individual files |
|  Purple | fc-edg<br>-ai  | Local-Only / AI-Sensitive — content the on-device AI flagged as never permitted to leave the device. Used for the personal Secure Vault and any folder marked "Local Only". | Secure Vault, Sensitive Docs, Local Only           |

These same tokens drive the rest of the visual system — the **chip badges** ("Restricted", "Confidential", "Internal", "Public") on file rows, the **border-left** strips on folder list items, the **icon background tints**, and the per-classification color theming inside the File Viewer's Security Intelligence Panel. So the color on the folder edge is a quick scan of "what's the highest-classified content in here" — matching the badge you'll see when you tap in.





## 8. iPad Files Experience

### Split-View Workspace

#### Left Panel

Repository tree and navigation

#### Center Panel

File browser and search results

#### Right Panel

Preview / metadata / governance / AI insights

### Touch Interactions

- Drag-and-drop
- Long press context menus
- Swipe actions
- Multi-select gestures
- Keyboard + trackpad support





## 9. Secure File Viewer

### Purpose

Protected runtime document experience.

### Supported Content

- PDF
- DOCX
- XLSX
- TXT
- Images
- Future multimedia support

### Runtime Protections

- Screenshot restriction
- Screen recording detection
- Watermark overlays
- Copy prevention
- Background blur
- Secure rendering isolation

### AI Intelligence Layer

#### Ambient Analysis

- Sensitivity detection
- Risk scoring
- Compliance mapping
- Policy reasoning

#### Dynamic Enforcement

AI may:

- Disable sharing
- Force view-only mode
- Require biometric reauthentication





- Escalate policy enforcement

## 10. AI Workspace

### Purpose

Unified AI-assisted secure productivity environment.

### Multi-Panel Adaptive Layout

#### Mobile

Single focused workflow

#### iPad

Resizable productivity panels

#### Desktop

Full multi-pane workspace

### Panels

- AI Chat
- Sources
- Governance
- Classification
- Suggested Actions
- Output Preview
- Audit Context

### AI Capabilities

- Summarization
- Classification





- Risk explanation
- Data lineage analysis
- Governance recommendations
- Secure workflow orchestration

## 11. Secure Email Workspace

### Inbox Features

- Secure message rendering
- Attachment governance
- Sensitivity indicators
- AI threat/risk analysis

### Compose Experience

AI continuously evaluates:

- Recipients
- Attachments
- Content
- Compliance posture

### Enforcement Examples

- Force encryption
- Restrict recipients
- Block sending
- Require approval
- Recommend secure sharing





## 12. Sharing Center

### Purpose

Unified external access governance.

### Features

- Active shares
- External recipients
- Secure links
- Expiration controls
- Revocation management
- Share analytics

### AI Risk Signals

- Sensitive recipient detection
- Geographic risk
- Compliance conflicts
- Oversharing patterns

## 13. Notifications & Events

### Categories

- Security risks
- Policy violations
- Sync conflicts
- Access changes
- AI recommendations
- Repository health

### User Experience Goals





- Minimal interruption
- High signal-to-noise ratio
- Prioritized alerts
- Human-readable explanations

## 14. Enterprise Governance Workspace

### Role-Based Access

Visible only to:

- Enterprise admins
- Governance operators
- Security teams

### Functional Areas

#### Policy Management

- Sharing rules
- DLP controls
- Runtime restrictions
- Data residency

#### AI Governance

- Confidence thresholds
- Escalation logic
- Human override policies
- AI explainability settings

#### Device Trust

- Compliance validation
- Runtime health
- Device risk posture







## **Audit & Activity**

- File access
- Sharing events
- AI decisions
- Classification changes
- Trust events

# **15. Settings Architecture**

## **Categories**

### **Account**

- Identity
- SSO
- Workspace membership

### **Security**

- Face ID
- App lock
- Session timeout

### **AI Governance**

- Privacy preferences
- AI explainability
- Local vs cloud AI usage

### **Offline Storage**

- Cache management
- Offline repository selection





## Repository Connections

- Add/remove repositories
- Reauthentication
- Sync configuration

## Notifications

- Risk sensitivity
- Alert preferences

## Appearance & Workspace

- Mobile/iPad/Desktop mode
- Theme preferences
- Layout density





## 16. AI Interaction Model

### Ambient AI Philosophy

AI should remain:

- Quiet
- Contextual
- Non-intrusive
- Explainable

### Explicit AI Interaction Triggers

Only surface prominently when:

- User decision required
- Risk identified
- Policy conflict exists
- Compliance action needed

### AI Communication Style

- Human-readable
- Minimal technical jargon
- Action-oriented
- Transparent reasoning





## 17. Adaptive Layout System

### Responsive Breakpoints

| Mode           | Layout Style     |
|----------------|------------------|
| Mobile         | Single-column    |
| iPad Portrait  | Adaptive stacked |
| iPad Landscape | Split-pane       |
| Desktop        | Multi-workspace  |

### Requirements

- Shared adaptive components
- No duplicated screens
- Runtime layout transitions
- Orientation-aware rendering

## 18. Security Runtime Requirements

### Mandatory Protections

- Biometric unlock
- App blur on background
- Secure enclave usage
- Token encryption
- Runtime integrity validation





## High-Risk Escalation

Sensitive content may dynamically:

- Increase watermarking
- Disable exports
- Require step-up authentication
- Restrict offline storage

# 19. Offline-First Architecture

## Requirements

- Local encrypted cache
- Secure offline editing
- Deferred sync
- Conflict resolution
- Offline AI classification

## Sync Engine

Supports:

- Background sync
- Policy-aware synchronization
- Intelligent bandwidth usage
- Conflict remediation





## 20. Future Expansion Modules

### Planned Capabilities

- Secure team collaboration
- Secure chat
- AI governance analytics
- Workflow automation
- Cross-device workspace continuity
- Secure browser mode
- Enterprise DLP dashboards
- Edge AI policy enforcement
- AI-driven operational copilots

## 21. Design System Requirements

### Visual Design

- Native iOS/iPadOS aesthetics
- Minimalist enterprise UI
- Flat iconography
- Soft depth hierarchy
- Motion-driven context transitions

### Interaction Design

- Touch-first
- Gesture-aware
- Keyboard compatible
- Accessibility compliant





## Accessibility

- Dynamic text scaling
- VoiceOver compatibility
- WCAG compliance
- High contrast support

# 22. QA & Validation Requirements

## Test Modes

- Mobile portrait
- Mobile landscape
- iPad portrait
- iPad landscape
- Split-screen multitasking
- Stage Manager
- External display mode
- Desktop web

## Validation Areas

- Runtime security
- Gesture interactions
- Offline workflows
- AI governance enforcement
- Mode switching continuity
- Performance under sync load
- Cross-repository behavior



# Technical considerations





# XQ Secure Workspaces AI

## Technical Architecture & Mobile Application Specification

iOS / iPadOS Enterprise Zero Trust Data Platform

## 1. System Overview

### Product Objective

XQ Secure Workspaces is an AI-native Zero Trust mobile platform for secure file access, secure communications, governed collaboration, and persistent data protection across enterprise and sovereign environments.

The application provides:

- AI driven UX
- Secure repository access
- AI-driven governance
- Persistent encryption
- Dynamic policy enforcement
- Cross-platform extensibility
- Modular integration architecture
- Offline-first protected workspace capabilities

## 2. Core Architectural Principles

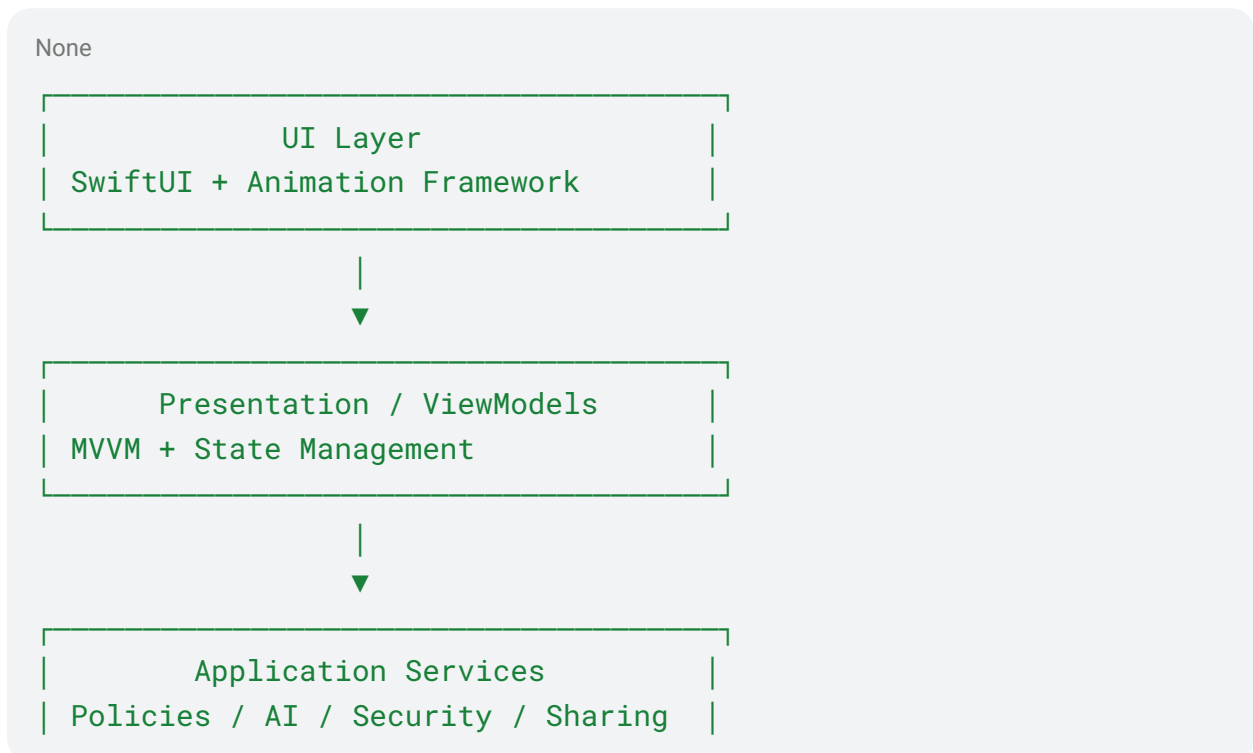
| Principle                     | Description                               |
|-------------------------------|-------------------------------------------|
| Interface-Driven Architecture | All services abstracted behind interfaces |

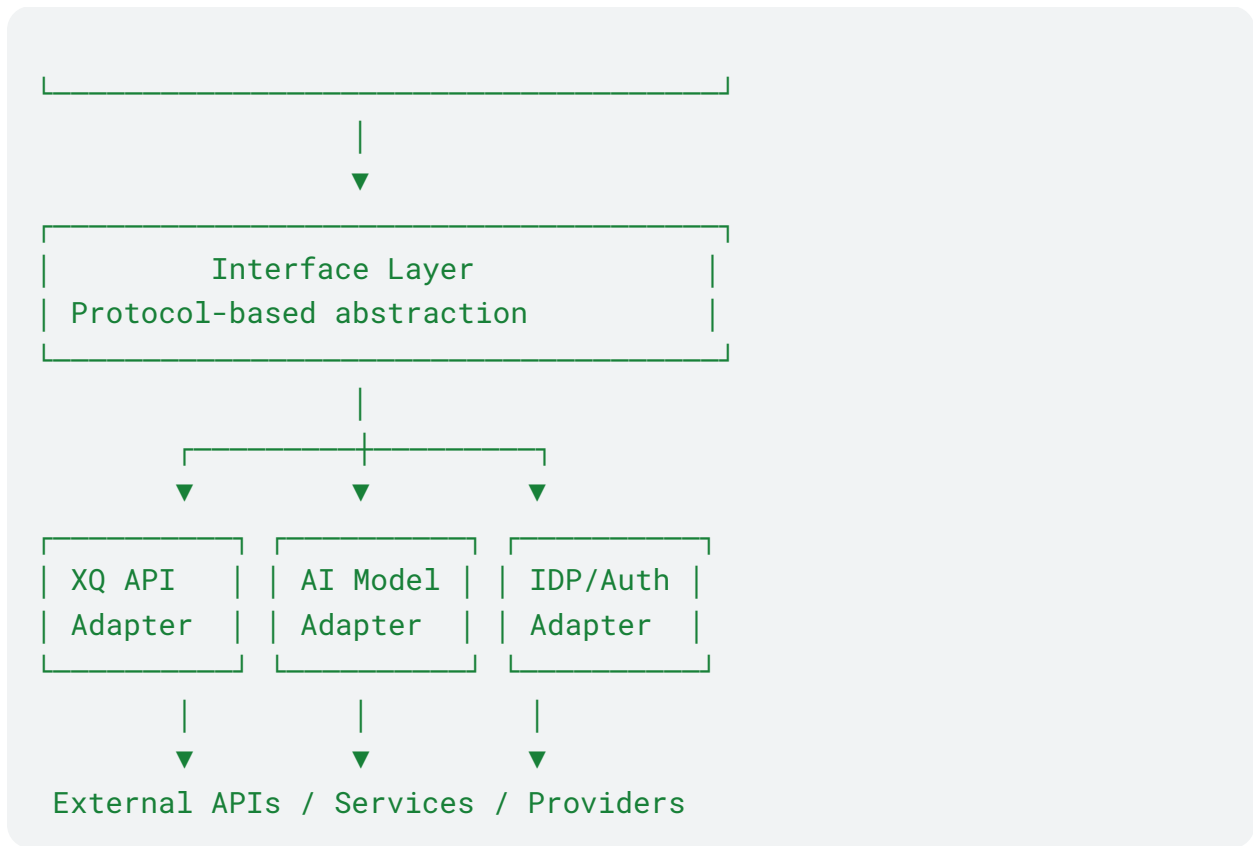




|                                  |                                                                                             |
|----------------------------------|---------------------------------------------------------------------------------------------|
| No Direct API Coupling           | No business logic directly calls external APIs                                              |
| AI Provider Agnostic             | Interchangeable AI models/providers. Local model at the edge. (on Device IOS, LLama, Gemma) |
| Zero Trust Data-Centric Security | Protection follows data                                                                     |
| Offline-First                    | Core functions operate without connectivity                                                 |
| Policy-Driven Runtime            | Runtime behavior dynamically controlled                                                     |
| Cross-Platform Core              | Shared architecture for iOS/Android/Desktop                                                 |
| Localization-Native              | All UI text externally managed                                                              |
| Adaptive UI                      | Layout auto-adjusts for localization/accessibility                                          |
| Enterprise Extensible            | Modular enterprise integrations                                                             |

### 3. High-Level Architecture





## 4. Mobile Platform Stack

### iOS Technologies

| Layer                | Technology                 |
|----------------------|----------------------------|
| UI                   | SwiftUI                    |
| State                | Combine / Observable       |
| Architecture         | MVVM-C                     |
| Dependency Injection | Resolver / Factory         |
| Persistence          | CoreData / SQLite          |
| Encryption           | Secure Enclave + CryptoKit |





|                  |                              |
|------------------|------------------------------|
| Background Tasks | BGTaskScheduler              |
| Networking       | URLSession                   |
| Offline Sync     | Custom sync engine           |
| Logging          | Unified logging framework    |
| Animations       | SwiftUI + Core Animation     |
| AI Runtime       | CoreML / ONNX Runtime        |
| File Viewer      | QuickLook + Custom Renderers |

## 5. Interface-Based Architecture

### Requirement

No direct API calls allowed anywhere in business logic.

### 5.1 Interface Standards

All integrations must use interfaces.

Example:

Swift

```
protocol RepositoryProvider {  
    func authenticate()  
    func listFiles(path: String)  
    func downloadFile(id: String)  
    func uploadFile(file: SecureFile)  
}
```

Implementations:





- SharePointProvider
- GoogleDriveProvider
- SMBProvider
- LocalVaultProvider
- XQ API
- OAUTH

Business logic never references provider implementations directly.

## 5.2 Dependency Injection

All providers resolved dynamically.

Swift

```
class RepositoryService {  
    init(provider: RepositoryProvider)  
}
```

Benefits:

- Testability
- Mock support
- Runtime provider swapping
- Version compatibility
- Enterprise customization





## 6. XQ API Abstraction Layer

### Requirement

Support multiple XQ API versions simultaneously.

### 6.1 XQ API Interface

Swift

```
protocol XQSecureAPI {  
    func encrypt()  
    func decrypt()  
    func applyPolicy()  
    func revokeAccess()  
    func validateAccess()  
}
```

XQ API - <https://xq.stoplight.io/>

XQ SDK - <https://github.com/XQ-Message-Inc/>

### 6.2 Version Adapters

None

XQAPIv1Adapter

XQAPIv2Adapter

XQAPIv3Adapter

Version resolution:

- Dynamic capability negotiation
- Backward compatibility support
- Feature flag mapping





## 6.3 API Gateway Layer

All network traffic routed through:

- Request normalization
  - Retry orchestration
  - Token refresh
  - Telemetry
  - Audit logging
  - Security validation
- 

## 7. AI Architecture

### 7.1 AI Provider Abstraction

Swift

```
protocol AIProvider {  
    func classify(document: Data)  
    func generateSummary(text: String)  
    func evaluateRisk(context: RiskContext)  
}
```

Supported Providers:

- OpenAI
  - Anthropic
  - AWS Bedrock
  - Azure OpenAI
  - Local CoreML Models
  - ONNX Models
  - Enterprise custom models
- 

### 7.2 AI Model Switching





Runtime configurable:

- Per tenant
- Per policy
- Per data classification
- Offline fallback

Example:

- Local model for CUI
- Cloud model for summarization

## 7.3 AI Governance Engine

Functions:

- Classification
  - Sensitivity scoring
  - Risk assessment
  - Policy recommendations
  - Behavioral analysis
  - Sharing validation
- 
- This can borrow from the work we've already done here

## 8. Fuzzy Logic Policy Engine







## Purpose

Policies are adaptive rather than binary.

Instead of:

None

ALLOW / DENY

System evaluates:

- Risk score
- Device trust
- User behavior
- Location
- Sensitivity
- Sharing patterns
- Time
- Session anomalies

## 8.1 Example

None

IF:

External User

AND Sensitivity = High

AND Device Trust = Medium

THEN:

Allow View Only

Disable Download

Require MFA Every 5 Minutes

Enable Heavy Watermark





## 8.2 Policy Engine Architecture

None

Policy Inputs



Risk Evaluation Engine



Fuzzy Logic Processor



Runtime Controls



XQ Enforcement Layer

## 9. Identity & Group Model

### 9.1 Identity Providers

Supported:

- Microsoft Entra ID
- Okta
- Google Workspace
- Ping
- AWS IAM Identity Center

### 9.2 Gmail Group Logic





Requirement:  
Gmail identity acts as group container.

Example:

None

`project-team@gmail.com`

Automatically becomes:

- Shared secure workspace
- Shared folder boundary
- Shared governance container

## 9.3 Dynamic Group Policies

Groups inherit:

- Labels
- Access policies
- Sharing restrictions
- Retention rules
- Recipients

## 10. Shared Folder Collaboration Model





## Invite-Based Workspace Creation

User Flow:

1. User invites participants
2. App creates encrypted workspace container
3. Policies applied automatically
4. Shared folder established
5. Dynamic membership maintained

---

## 10.1 Shared Workspace Characteristics

| Capability            | Description               |
|-----------------------|---------------------------|
| Persistent encryption | Data remains protected    |
| Membership governance | Dynamic access            |
| Revocation            | Immediate removal         |
| Activity logging      | Full auditability         |
| Offline sync          | Controlled secure cache   |
| AI governance         | Continuous classification |

---

## 11. Localization & Language System

### 11.1 Multi-Language Architecture

Requirement:

All UI text externalized.

---





## 11.2 Localization Files

None

```
/en/messages.json  
/fr/messages.json  
/de/messages.json  
/jp/messages.json
```

Example:

JSON

```
{  
  "home.title": "Home",  
  "files.import": "Import File",  
  "risk.warning": "Sensitive file detected"  
}
```

---

## 11.3 Runtime Language Switching

Features:

- Dynamic reload
- No app restart required
- Tenant-specific language packs
- OTA translation updates

---

## 11.4 Adaptive UI

Requirements:

- Auto-resizing buttons
- Flexible containers
- RTL support





- Dynamic type scaling
- Accessibility-first layouts

---

## 12. Animation System

### Requirement

Every UX element must animate.

---

### 12.1 Animation Categories

| Type         | Requirement         |
|--------------|---------------------|
| Entry        | Fade + motion       |
| Exit         | Scale + fade        |
| Click        | Spring response     |
| Navigation   | Smooth transitions  |
| Risk Alerts  | Contextual emphasis |
| File Actions | Gesture animation   |

### 12.2 Animation Framework

Use:

- SwiftUI animations
- Core Animation
- Motion interpolation engine





- Research the best UI framework for centralized Design control. Prioritize a cross platform module.

Performance requirements:

- 60 FPS minimum
- Reduced motion accessibility support

## 12.3 UI Guidance

For a modern iOS-first app that also needs strong Android consistency, the top choices in 2026 are:

| Framework /<br>Module Stack                            | Native<br>Feel | Cross-Platform<br>Consistency | Performance | Enterprise<br>Suitability | Best Use<br>Case                               |
|--------------------------------------------------------|----------------|-------------------------------|-------------|---------------------------|------------------------------------------------|
| Kotlin Multiplatform<br>+ SwiftUI +<br>Jetpack Compose | Excellent      | High (logic<br>shared)        | Native      | Excellent                 | Enterprise,<br>security, AI,<br>regulated apps |
| Flutter                                                | Very<br>Good   | Excellent                     | Near-native | Very Good                 | Pixel-perfect<br>UX/UI<br>consistency          |
| React Native                                           | Good           | Good                          | Near-native | Good                      | Fast iteration,<br>JS teams                    |
| Compose<br>Multiplatform                               | Very<br>Good   | Excellent                     | Near-native | Emerging                  | Kotlin-centric<br>shared UI                    |

Based on your requirements from prior discussions — enterprise security, modular architecture, interchangeable AI models, policy enforcement, adaptive UI, multilingual support, iPad + iPhone support, and likely long lifecycle maintenance — the strongest architecture is:





# Recommended Architecture

## 1. Core UX/UI Stack

### iOS

- SwiftUI

### Android

- Jetpack Compose

### Shared Layer

- Kotlin Multiplatform

This gives:

- True native UX on each platform
- Shared business logic
- Shared networking
- Shared policy engine
- Shared security layer
- Shared AI orchestration
- Shared localization framework
- Shared design tokens

This is increasingly considered the enterprise-grade approach for regulated/security-sensitive mobile applications.

## Why This Is Better Than Flutter

Flutter is excellent for UI consistency, but your app requirements lean heavily toward:

- security
- native OS integrations
- policy enforcement
- background processing
- device posture







- identity providers
- external SDK integrations
- secure storage
- MDM compatibility
- AI model abstraction
- enterprise compliance

These are areas where native platform access matters.

Kotlin Multiplatform lets you:

- share 60–80% of code
- maintain fully native UI
- avoid bridge performance issues
- integrate deeply with iOS and Android security APIs

This is especially important for:

- Zero Trust enforcement
- secure enclaves
- device attestation
- offline policy evaluation
- AI model switching
- encrypted local storage

## Best UX/UI Modules To Standardize

### Design System Layer

Use:

- Material 3 tokens internally
- Apple Human Interface Guidelines on iOS
- Shared design tokens via KMP

Recommended structure:





None

### Shared Design Tokens

└─ colors

└─ typography

└─ spacing

└─ iconography

└─ motion

└─ accessibility

└─ responsive breakpoints

## Best Native UI Modules

### Navigation

#### iOS

- UINavigationController
- SplitView for iPad

#### Android

- Navigation Compose

### State Management

Recommended:

- MVVM + unidirectional data flow





Use:

- StateFlow
- Combine
- Shared ViewModels via KMP

# Best Cross-Platform UI Component Systems

## Best Overall

### Compose Multiplatform

Best future-facing option if you later want:

- desktop
- web
- Android Automotive
- embedded AI terminals

But:

- iOS support is still maturing

## Best Stable Enterprise Choice

### SwiftUI + Compose + KMP

This is currently the safest long-term architecture.





# Recommended Modular UX/UI Architecture

None

## Presentation Layer

└─ Native UI

|     └─ SwiftUI

|     └─ Jetpack Compose

|

└─ Shared Design System

|

└─ Shared State Layer

|

└─ Shared Policy Engine

|

└─ Shared AI Interface

|

└─ Shared Security Framework

|

└─ Shared Localization Engine





# Important UX/UI Requirements For Your App

Given your previous requirements, your UI framework should support:

## Dynamic Policy-Controlled UI

Example:

- Hide features by RBAC/ABAC
- Dynamic labels
- Geofenced UI
- Tenant-based layouts

KMP handles this very well.

## Externalized Localization

Use:

```
JSON
/localization/en.json

/localization/fr.json

/localization/ar.json
```

Do NOT hardcode strings.

## Adaptive Layout System

Must support:

- iPhone
- iPad
- foldables
- Android tablets





- external monitors

SwiftUI + Compose are best-in-class here.

## Accessibility

Need:

- Dynamic text
- VoiceOver/TalkBack
- keyboard navigation
- high contrast
- RTL layouts

Native UI frameworks handle accessibility significantly better than Flutter or React Native.

# Final Recommendation

## Best Overall Architecture

### Kotlin Multiplatform

- 

### SwiftUI

- 

### Jetpack Compose

This gives you:

- native-grade UX
- shared business/security logic
- Android/iOS consistency
- enterprise scalability
- best long-term maintainability
- strong AI integration flexibility





- best security posture

Flutter is the second-best choice if:

- maximum visual consistency matters more than native fidelity.

React Native is best only if:

- your organization is heavily JavaScript/TypeScript-centric.

## 13. Resource Control System

### Purpose

Prevent excessive:

- Battery drain
- AI processing
- Memory consumption
- Network utilization

### 13.1 Adaptive Resource Policies

Dynamic controls:

- AI throttling
- Sync batching
- Background limits
- GPU usage caps
- Cache pruning





## 13.2 Device-Aware Scaling

Example:

Older devices:

- Reduced animation complexity
- Smaller AI models
- Limited background tasks

## 14. Secure File System

### 14.1 File Protection Model

All imported files:

- Encrypted immediately
- Bound to XQ policy metadata
- Stored in secure sandbox
- Never exposed directly to iOS filesystem

### 14.2 Runtime Controls

Protected Viewer:

- Screenshot blocking
- Screen recording detection
- Clipboard prevention
- Background blur
- Watermark overlays







## 14.3 Offline Storage

Offline files:

- Time-limited
- Policy-bound
- Auto-expiring
- Remote revocable

## 15. Integration Framework

### 15.1 Unified Integration Interface

Swift

```
protocol ExternalIntegration {  
    func authenticate()  
    func sync()  
    func query()  
    func upload()  
}
```

### 15.2 Supported Integrations

#### Microsoft

- SharePoint
- OneDrive
- Teams
- Exchange
- Entra ID





## Google

- Gmail
- Google Drive
- Google Workspace

## Additional

- Box
- Dropbox
- SMB
- AWS S3
- Local storage

# 16. Auto Update / Force Update System

## 16.1 Update Modes

| Mode          | Description                       |
|---------------|-----------------------------------|
| Silent Update | Background content/config updates |
| Soft Update   | Recommended app upgrade           |
| Force Update  | Mandatory security update         |

## 16.2 Dynamic Configuration

Remote config controls:

- Feature flags
- Policy updates
- AI model rollout





- Kill switches
- Version gating

## 17. Cross-Platform Strategy

### 17.1 Shared Architecture

Core shared concepts:

- APIs
- Policy engine
- AI abstraction
- Sync logic
- Security model

### 17.2 Future Platform Support

| Platform | Strategy                       |
|----------|--------------------------------|
| Android  | Kotlin equivalent architecture |
| macOS    | Native Catalyst                |
| Windows  | MAUI/Native                    |
| Web      | Secure browser workspace       |

## 18. Security Architecture





## 18.1 Core Security Controls

| Control             | Description           |
|---------------------|-----------------------|
| Secure Enclave      | Key storage           |
| Certificate Pinning | API security          |
| Runtime integrity   | Tamper detection      |
| Jailbreak detection | Compromise prevention |
| Device posture      | Trust scoring         |
| Secure cache        | Encrypted storage     |

## 18.2 Zero Trust Runtime

Continuous evaluation:

- Identity
- Device
- Session
- Location
- Behavior
- Classification
- Network posture
- version

## 19. Governance Architecture

### 19.1 Policy Types





| Policy         | Description         |
|----------------|---------------------|
| Classification | Data sensitivity    |
| Sharing        | External access     |
| Retention      | Expiration          |
| Device         | Trust requirements  |
| Geographic     | Region restrictions |
| Runtime        | Viewer protections  |

## 19.2 Label Architecture

Labels support:

- Follow MIPS
- Hierarchical inheritance
- Multi-label assignment
- AI confidence scoring
- Dynamic reclassification

## 20. Offline-First Sync Architecture

### Principles

- Local-first operations
- Eventual consistency
- Conflict resolution
- Secure sync queue





## 20.1 Sync Engine

Required for local groups

Capabilities:

- Delta sync
- Retry queue
- Background sync
- Policy-aware sync
- Selective offline availability

## 21. Enterprise Admin Framework

### Capabilities

#### Governance

- Policy authoring
- AI threshold tuning
- Group administration
- Access management

#### Audit

- File lineage





- User behavior
- Sharing events
- Device trust history

---

## 22. Observability & Telemetry

### Telemetry Categories

| Category | Description           |
|----------|-----------------------|
| Security | Threat events         |
| UX       | User interactions     |
| AI       | Model performance     |
| Sync     | Repository health     |
| Runtime  | Device<br>performance |

All telemetry:

- Policy governed
- Privacy aware
- Tenant configurable

---

## 23. Recommended Repository Structure

None

App/

├─ Core/

├─ UI/

├─ ViewModels/

├─ Services/





```
|— Interfaces/  
|— Providers/  
|— AI/  
|— Security/  
|— Policies/  
|— Localization/  
|— Animations/  
|— Sync/  
|— Integrations/  
|— Storage/  
|— Utilities/
```

## 24. Recommended Development Phases

### Phase 1

- Authentication
- File browser
- Secure viewer
- XQ encryption
- SharePoint integration
- Offline cache

### Phase 2

- AI classification
- Dynamic policy engine
- Secure sharing
- Gmail group workspaces

### Phase 3







- Advanced governance
- AI copilots
- Multi-model orchestration
- Cross-platform sync

## Phase 4

- Autonomous governance agents
- AI collaboration
- Enterprise analytics
- Secure workflow automation



# Security

# iOS App Security Best Practices — Code-Level Requirements

## 1. Secure Authentication & Identity

### Requirements

- Use platform-native authentication frameworks where possible.
- Never store passwords or tokens in plaintext.
- Support modern authentication:
  - OAuth 2.1
  - OpenID Connect (OIDC)
  - SAML (enterprise)
  - Passkeys/WebAuthn
- Enforce MFA for privileged operations.
- Implement token expiration and rotation.
- Use short-lived access tokens and refresh tokens.
- Prevent session fixation and replay attacks.

### Code-Level Best Practices

- Store credentials only in:
  - Keychain
  - Secure Enclave
- Use:
  - `LAContext`
  - `AuthenticationServices`
  - `CryptoKit`
- Require biometric revalidation before:
  - Export
  - File access
  - Admin actions
  - Key release

### Avoid

- `NSUserDefaults` for secrets
- Hardcoded API keys
- Embedded client secrets

- Local password caching

## 2. Keychain & Secure Storage

### Requirements

- All secrets must be encrypted at rest.
- Keys must be device-bound when possible.
- Use access control flags.

### Recommended Keychain Settings

- `kSecAttrAccessibleWhenUnlockedThisDeviceOnly`
- `biometryCurrentSet`
- `userPresence`

### Secure Storage Targets

- JWTs
- Refresh tokens
- Encryption keys
- Device identity
- Certificates

### Code-Level Controls

- Zeroize memory after secret usage.
- Prevent logs from containing secrets.
- Use ephemeral memory buffers for sensitive operations.

## 3. Cryptography Requirements

### Approved Algorithms

Prefer:

- AES-256-GCM
- ChaCha20-Poly1305
- Curve25519
- P-256
- SHA-256/384
- Ed25519

## **Use**

- CryptoKit
- CommonCrypto
- Secure Enclave APIs

## **Avoid**

- MD5
- SHA1
- ECB mode
- DES/3DES
- Custom cryptography

## **Key Requirements**

- Rotate keys regularly
- Separate KEK and DEK
- Support remote revocation
- Never export private keys if avoidable

# **4. Transport Security**

## **Requirements**

- TLS 1.2 minimum
- Prefer TLS 1.3
- Enforce ATS (App Transport Security)
- Use certificate pinning for high-security apps

## **Implement**

- Public key pinning
- Trust evaluation
- Revocation checking

## Avoid

- `NSAllowsArbitraryLoads = true`
- Disabling certificate validation
- Self-signed cert trust bypasses

## Recommended APIs

- `URLSession`
- `Network.framework`

# 5. Data Protection

## Requirements

- Encrypt sensitive local files.
- Apply iOS Data Protection classes.
- Use file-level encryption.

## Use

Swift

```
.completeFileProtection  
.completeFileProtectionUntilFirstUserAuthentication
```

## Sensitive Data Controls

Protect:

- Cached files
- AI prompts
- Attachments
- Logs
- Screenshots

- Clipboard data

### **Additional Controls**

- Clear clipboard automatically
- Prevent screenshots where appropriate
- Blur app in app switcher

## **6. Secure API Design**

### **Requirements**

- Validate all server responses.
- Implement schema validation.
- Use signed requests where possible.

### **API Security Controls**

- Nonce validation
- Timestamp validation
- Replay prevention
- Rate limiting
- Request signing
- Device attestation

### **Recommended**

- Mutual TLS (mTLS)
- Token binding
- Per-device identity certificates

## **7. Jailbreak & Runtime Protection**

### **Requirements**

Apps handling sensitive data should detect:

- Jailbreak/root indicators
- Dynamic instrumentation
- Runtime hooking
- Debuggers
- Emulators

## **Detect**

- Frida
- Cydia
- dylib injection
- ptrace bypass attempts

## **Runtime Defenses**

- Code obfuscation
- Integrity verification
- Binary hashing
- Anti-tamper checks

## **Important**

Jailbreak detection should:

- Increase risk score
- Reduce functionality
- Restrict sensitive operations

Avoid immediate app crashes unless policy requires it.

# **8. Secure Coding Standards**

## **Requirements**

Follow:

- OWASP MASVS
- OWASP MSTG



- CERT Secure Coding
- Apple Secure Coding Guide

## Input Validation

Validate:

- URLs
- JSON
- File uploads
- Deep links
- Intents
- AI prompt content

## Avoid

- Unsafe deserialization
- String interpolation in queries
- Dynamic code execution
- Untrusted WebViews

# 9. WebView Security

## Requirements

Use `WKWebView` only.

## Disable

- Arbitrary JavaScript bridges
- File URL access
- Mixed content

## Restrict

- Navigation domains
- Cookies
- Local storage
- Cross-origin requests

## **Avoid**

- UIWebView
- Eval-like JS execution
- Loading untrusted HTML

# **10. Dependency & Supply Chain Security**

## **Requirements**

- Maintain SBOM
- Pin dependency versions
- Review third-party SDKs

## **Validate**

- SDK permissions
- Telemetry collection
- Data sharing
- Embedded trackers

## **Security Scanning**

Use:

- SAST
- SCA
- Secret scanning
- Dependency vulnerability scanning

## **Avoid**

- Abandoned libraries
- Excessive analytics SDKs
- Ad SDKs in sensitive apps

# **11. Logging & Telemetry**

## Requirements

Logs must never contain:

- Tokens
- Keys
- PHI
- PII
- Secrets
- Message contents

## Implement

- Structured logging
- Redaction
- Secure log transport

## Use

- Unified logging (`os_log`)
- Privacy annotations

Example:

Swift

```
Logger.security.info("User authenticated: \(userID, privacy:  
.private)")
```

# 12. Privacy & Permissions

## Requirements

Use least privilege access.

## Request Permissions Only When Needed

- Camera
- Photos
- Contacts

- Bluetooth
- Microphone
- Location

### **Additional Controls**

- Explain permission usage clearly
- Detect revoked permissions
- Handle denial securely

## **13. Memory & Process Security**

### **Requirements**

- Prevent sensitive data persistence in memory.
- Minimize object lifetime.

### **Use**

- Secure memory handling
- Autorelease pool discipline
- Background clearing

### **Protect Against**

- Memory scraping
- Screen recording
- Snapshot leakage

## **14. AI/LLM-Specific Security Requirements**

For AI-enabled applications:

### **Requirements**

- Sanitize prompts

- Enforce policy before model submission
- Apply data labeling
- Prevent prompt injection

## **Implement**

- Output filtering
- DLP scanning
- Context isolation
- Tenant isolation
- Prompt provenance tracking

## **Protect**

- RAG sources
- Embeddings
- Model credentials
- AI conversation history

## **Recommended**

- Local policy enforcement layer
- Model abstraction interface
- Secure tool execution sandbox

# **15. Enterprise / Zero Trust Requirements**

## **Requirements**

Support:

- RBAC
- ABAC
- Conditional access
- Device posture validation

## **Integrate With**

- MDM/UEM
- Identity Providers

- SIEM
- EDR/XDR
- Data governance engines

## **Enforce**

- Geofencing
- Time-based access
- Risk-based access
- Remote wipe
- Remote key revocation

# **16. Build Pipeline Security**

## **Requirements**

Protect:

- Signing certificates
- CI/CD secrets
- Provisioning profiles

## **Implement**

- Reproducible builds
- Signed artifacts
- Branch protections
- Mandatory reviews

## **Use**

- Secret scanning
- Binary analysis
- Dependency verification

# **17. App Store & Provisioning Security**

## Requirements

- Use hardened entitlements.
- Remove unused capabilities.

## Validate

- Associated domains
- Push notification usage
- Background modes
- Keychain sharing

## Avoid

- Excess entitlements
- Debug symbols in release builds
- Test endpoints in production

# 18. Recommended Security Frameworks & Standards

## Apple Frameworks

- CryptoKit
- Network.framework
- AuthenticationServices
- LocalAuthentication
- DeviceCheck
- App Attest

## Security Standards

- OWASP MASVS
- NIST SP 800-63
- NIST Zero Trust Architecture
- FIPS 140-3 validated cryptography
- SOC 2
- HIPAA

- GDPR
  - CMMC
  - FedRAMP
- 

## 19. Recommended Architecture Pattern

For enterprise-grade secure iOS apps:

None

UI Layer



Policy Enforcement Layer



Secure Service Layer



Encrypted Data Layer



Trusted API Gateway



Zero Trust Backend

### Core Principles

- No direct API access from UI
- Centralized authorization
- Data-centric security
- Externalized policy enforcement
- Cryptographic identity everywhere
- Assume breach architecture



# Sitemap



# XQ Secure Workspaces

## Sitemap

### Top-Level Navigation Model

#### Device Mode Switcher

- Floating control positioned bottom-left
- Modes:
  - Mobile
  - iPad
  - Desktop
- Stored in localStorage:
  - `xq.deviceMode`

#### Primary Navigation Structure

##### Top Header

- Large title area:
  - Files
  - Messages
  - Now
  - Profile
- Top-right avatar [BW]
  - Opens Profile

##### Navigation Layout

##### Mobile

- Bottom tab bar
  - Files
  - Messages
  - Now





iPad/Desktop

- Left sidebar rail
  - Files
  - Messages
  - Now
  - Bottom-anchored Profile avatar

## Device Layouts

- Mobile:
  - 390 × 844
  - Bottom navigation
- iPad:
  - 1024 × 768
  - 236px sidebar
- Desktop:
  - 1440 × 900
  - 264px sidebar
  - 4-column folder grid

# 1. Onboarding Paths

## Initial Flow

None

Splash



Welcome

## Welcome Screen Options

- Start for Free
  - Personal onboarding
- Connect Enterprise Workspace
  - Enterprise onboarding





## 1.1 Personal / Local-First Flow

```
None
Welcome
  ↓
Repository Setup
  ↓
Permissions
  ↓
Notifications (Now)
```

### Notes

- Workspace selection is skipped
- Back navigation controlled by:
  - `xq.onbPath`

## 1.2 Enterprise (SSO) Flow

```
None
Welcome
  ↓
Enterprise Authentication
  ↓
Workspace Selection
  ↓
Permissions
  ↓
Notifications (Now)
```





## 2. Now Screen (s-notifications)

### Purpose

Primary home surface.

### Sub-Tabs

1. Overview
2. Ask (AI Assistant)
3. Activity
4. Security

### Overview Content

- Trust status pill
- Today event feed
- Needs Your Attention cards
- Suggested AI actions

### Navigation Targets

- Files
- Email
- Sharing
- Received Shares
- Group Invites
- Groups
- Profile





## Ask (AI Assistant)

### Features

- XQ AI Assistant header
- Capability cards
- Suggested prompts
- AI conversation stream
- Sticky chat input

## 3. Files (**s-file-browser**)

### Header Actions

- Title: Files
- **+** button
  - Create / Import overlay
- Risk button
  - Risk Overview overlay
- More menu
- Profile avatar

### Search

- Routes to:
  - Now → Ask

### Files Sub-Tabs

1. Folders
2. Recent
3. Shared
4. Vault





## 3.1 Create / Import Menu

### Create New

- Blank Document
- Memo
- Report
- Meeting Agenda
- Structured Form

### Import Existing

- Import File
  - Opens AI classification flow

## 3.2 File Destinations

| File Type         | Destination     |
|-------------------|-----------------|
| Editable document | Document Editor |
| View-only file    | File Viewer     |
| Folder            | Folder View     |

## 3.3 Files Child Screens

| Screen       | Purpose                 |
|--------------|-------------------------|
| Folder View  | Workspace folder detail |
| Folder Empty | Empty folder state      |





|                     |                                    |
|---------------------|------------------------------------|
| AI Organize         | AI folder organization             |
| AI Import           | Import + classification            |
| File Viewer         | Secure document viewer             |
| Doc Editor          | Native document editor             |
| File Risk Dashboard | Workspace risk analytics           |
| Data Lineage        | File provenance graph              |
| Screenshot Block    | Screenshot prevention interstitial |

## 3.4 File Viewer Structure

### Header Actions

- Back to Files
- Edit
- AI
- Share

### Components

- Edit-state strip
- Watermark
- File content area
- Security & Intelligence Panel

### SIP Sections

- Data Protection
- Risk / Threat Scan
- AI Summary
- Lineage / Provenance







## 3.5 Document Editor Structure

### Navigation

- Back to Files
- Presence row
- New Document pill
- Share
- Done

### Collaboration Features

- Collaborator avatars
- Track Changes
- Suggesting
- Comments
- Versions
- Request Approval

### Additional Features

- Classification banner
- View/Edit tabs
- Formatting toolbar
- AI Assist
- Editable content area





## 4. Messages (**s-email-inbox**)

### Header Actions

- Compose button
- Profile avatar

### Search

- Routes to:
  - Now → Ask

### Email Rows

- Open Email Thread

## 4.1 Message Screens

| Screen         | Purpose                   |
|----------------|---------------------------|
| Email Thread   | Threaded secure email     |
| Phishing Alert | Threat analysis           |
| Email Compose  | Secure compose experience |

## 4.2 Email Compose Structure

### Navigation

- Cancel
- New Secure Message
- Send





## Core Sections

- External recipient warning
- To / Cc / Bcc
- Subject
- Encryption controls
- Formatting toolbar
- AI controls
- Attachments

## AI Composition Drawer

### Features

- Help Me Write
- Rewrite styles:
  - Professional
  - Concise
  - Executive
  - Friendly
  - Technical
- Quick AI actions
- Streamed output
- Insert into draft
- Governance footer

## Additional Email Intelligence

- Tone analysis
- Commitments detection
- AI risk scoring
- Send confirmation sheet





## 5. Profile (s-profile)

### Purpose

Centralized account and workspace management.

### Main Sections

- Identity card
- Security Health
- Quick Actions
- Sticky subsection navigation

### Profile Subsections

| Section       | Features                          |
|---------------|-----------------------------------|
| General       | Theme, display mode, support      |
| Security      | MFA, biometrics, recovery         |
| Notifications | Push, email, SMS settings         |
| Integrations  | SharePoint, OneDrive, Slack, Okta |
| Workspace     | Tenant and compliance info        |
| Devices       | Active sessions and revocation    |
| Billing       | Plan and licensing                |
| Admin         | Policies and governance           |





## 6. Sharing System

### Sharing Screen

- Active shares
- Expiring shares
- Revocation controls

### Share Securely Flow

#### Step 1 — Recipients

- People
- Groups

#### Step 2 — AI Risk Analysis

#### Step 3 — Share Settings

##### AI Sharing Assistant

- Overexposure analysis
- Suggested expiry
- Resharing recommendations

##### Resharing Modes

- None
- Internal
- Group-Only
- Approval
- External-Prohibited
- Owner
- Policy
- Temporary

##### Additional Settings

- Expiry





- Screenshot detection
- Require XQ
- Audit logging
- Auto-grouping

#### **Step 4 — Confirmation**

#### **Step 5 — Admin Approval**

## **Additional Sharing Screens**

| Screen            | Purpose               |
|-------------------|-----------------------|
| Received<br>Share | Share acceptance flow |
| Group Invite      | Invitation management |

## **7. Workspaces / Groups**

| Screen              | Purpose                |
|---------------------|------------------------|
| Groups              | Workspace<br>dashboard |
| Workspace<br>Select | Workspace switcher     |
| Group Invite        | Invite acceptance      |





## 8. AI Surfaces

### Canonical AI Surface

- Now → Ask sub-tab

### AI Entry Points

- Files search
- Email search
- Email Compose AI
- File Viewer AI
- Doc Editor AI
- Share Assistant AI

## 9. Overlays & Bottom Sheets

| Overlay              | Purpose                 |
|----------------------|-------------------------|
| Risk Overlay         | Quick risk summary      |
| Create Menu          | Create/import chooser   |
| Insights Overlay     | Expanded security panel |
| Share Sheet          | Secure sharing workflow |
| Compose Send Sheet   | Send confirmation       |
| Agent Auth Sheet     | Enterprise consent      |
| Compose AI Drawer    | Inline AI composition   |
| File Viewer AI Panel | AI insights panel       |





## 10. `go()` Alias Routing

| Alias                          | Actual Destination |
|--------------------------------|--------------------|
| <code>s-home</code>            | Notifications      |
| <code>s-settings</code>        | Profile            |
| <code>s-policy</code>          | Profile → Admin    |
| <code>s-ai-assistant</code>    | Now → Ask          |
| <code>s-semantic-search</code> | Now → Ask          |

## 11. Persistent Local Storage State

| Key                            | Values                  | Purpose                 |
|--------------------------------|-------------------------|-------------------------|
| <code>xq.deviceMode</code>     | mobile / ipad / desktop | Layout selection        |
| <code>xq.notifMode</code>      | expanded / minimized    | Notification style      |
| <code>xq.profileSection</code> | subsection names        | Restore profile section |
| <code>xq.onbPath</code>        | personal / enterprise   | Onboarding state        |







## 12. Screen Inventory

### Active Screens

- AI Import
- AI Organize
- Data Lineage
- Doc Editor
- Email Compose
- Email Inbox
- Email Thread
- Enterprise Auth
- File Browser
- File Risk Dashboard
- File Viewer
- Folder Empty
- Folder View
- Group Invite
- Groups
- Notifications
- Permissions
- Phishing Alert
- Profile
- Received Share
- Repository Setup
- Screenshot Block
- Sharing
- Splash
- Welcome
- Workspace Select

### Counts

- 26 active screens
- 8 overlays / sheets
- 4 Now sub-tabs
- 4 Files sub-tabs



# Phased Development



# Phased Development

Here is a structured phased development and release plan based on the prioritized feature sets, user workflows, and technical roadmaps outlined in the XQ Secure Workspaces specification.

The strategy systematically builds the platform from a foundational secure container for files up to a fully autonomous, AI-governed enterprise ecosystem.

## Phase 1: Foundation & Secure File Vault

The initial phase focuses on establishing the secure app container, baseline encryption, and core file management workflows. This phase intentionally excludes complex features like real-time collaborative editing and separate admin web consoles to prioritize core security and local-first operations.

- **Secure File Vault:** Deployment of the primary protected workspace for storing, viewing, and organizing files.
- **Authentication & Access:** Implementation of basic user authentication workflows.
- **File Browser & Secure Viewer & Editor:** Introduction of the UI for navigating files and viewing them within a protected runtime environment.
- **XQ Encryption:** Implementation of persistent Zero Trust data protection and local-first encrypted storage.
- **SharePoint Integration:** Native connections to Microsoft SharePoint to allow streaming and accessing files without storing them on XQ infrastructure.
- **Offline Cache:** Enabling a secure local cache so users can operate on designated files without an active internet connection.

## Phase 2: Secure Email, Intelligence & Collaboration

With the secure file foundation in place, Phase 2 introduces communications, ambient AI intelligence, and active policy enforcement. The focus shifts to how data moves and how AI can automate compliance without user friction.

- **Secure Email Integration:** Integration of a secure email experience with protected attachment handling directly into the workspace.
- **AI Content Classification:** Deployment of on-device AI to continuously scan and classify documents and emails based on sensitivity.





- **Dynamic Policy Engine:** Activation of automated governance where the system applies encryption, restricts actions, or triggers alerts based on the AI's classification.
- **Secure Sharing Workflows:** Implementation of auditable, AI-assisted sharing controls that validate recipients and apply appropriate protections automatically.
- **Gmail Group Workspaces:** Establishing shared secure workspaces and folder boundaries utilizing Gmail identities as group containers.

## Phase 3: Secure Chat & Advanced AI Capabilities

Phase 3 completes the core unified workspace triad by adding chat capabilities and scales the platform for heavy enterprise use through advanced AI copilots and cross-platform syncing.

- **Secure Chat:** Rollout of private, secure messaging capabilities within the protected container.
- **Advanced Governance:** Expansion of enterprise administrative controls, allowing for centralized policy publishing, compliance visibility, and dynamic enforcement tuning.
- **AI Copilots:** Introduction of embedded AI composition assistance for drafting, summarizing, rewriting, and analyzing communications and documents.
- **Multi-Model Orchestration:** Deployment of the XQ AI Orchestration Layer to route requests between local CoreML/ONNX models and cloud providers (like OpenAI or Anthropic) based on data classification.
- **Cross-Platform Sync:** Implementation of synchronization capabilities to ensure secure workspace continuity across devices.

## Phase 4: Autonomous Operations & Enterprise Scale

The final phase matures the platform into a highly automated, fully governed enterprise ecosystem.

- **Autonomous Governance Agents:** Deployment of AI agents capable of making continuous, real-time security and compliance decisions with minimal human intervention.
- **AI Collaboration:** Enhanced features for AI-driven team collaboration and shared governance intelligence.
- **Enterprise Analytics:** Rollout of comprehensive audit tracking, data lineage graphs, and DLP (Data Loss Prevention) dashboards for security teams.





- **Secure Workflow Automation:** Automation of complex enterprise data workflows, ensuring data remains protected, portable, and policy-bound throughout its lifecycle.

